Decision Trees and Impurity Metrics

Decision Trees perform recursive binary partitioning of the feature space. Split selection chooses the feature and threshold that maximizes Information Gain or minimizes node impurity.

Metrics:
- Entropy: H(S) = - sum(p_i * log2(p_i)). Information Gain = H(parent) - weighted_sum(H(children)).
- Gini Impurity: Gini = 1 - sum(p_i^2). Computationally faster as it avoids logarithmic calculations.
Unconstrained decision trees grow deep leaves and overfit training data, necessitating pre-pruning (max_depth, min_samples_split) or cost-complexity pruning.

Random Forests and Gradient Boosting Ensembles

Ensemble methods combine multiple base estimators to achieve superior predictive accuracy:
1. Bagging (Bootstrap Aggregation): Trains diverse models on random bootstrap subsets of data and averages predictions. Random Forests enhance Bagging by also subsampling features at each split, decorrelating individual trees and reducing variance.
2. Boosting: Trains models sequentially, where each new estimator corrects residual errors made by preceding estimators. Gradient Boosted Decision Trees (GBDT) fit trees to the negative gradient of the loss function.

Validation Protocols and Bias-Variance Decomposition

Expected Generalization Error decomposes into: Error = Bias^2 + Variance + IrreducibleNoise.

To estimate out-of-sample generalization reliably:
- K-Fold Cross Validation partitions training data into K equal folds, training on K-1 folds and validating on the held-out fold across K iterations.
- Stratified K-Fold ensures each fold preserves the original class label distribution, essential for imbalanced classification tasks.